

Task 3-1 DML commands using clauses, operators and functions in queries

AIM:

To implement DML commands using clauses, operators and functions in queries

Data Manipulation Language

The DML is used to retrieve insert and modify database information. These commands will be used by all database users during the routine operation of the database.

DML commands

1. Insert into: This is used to add record into a relation

Syntax: Insert into <table name> (field 1, field 2, --- field n)
values (data 1, data 2, --- data n)

Ex: INSERT INTO Customer (SSN, Name, Email, Phone, Address)
VALUES ('CUST003', 'Michael', 'mike@gmail.com', '987651111', 'Chicago')

After inserting

SSN	Name	Email	Phone	Address
CUST001	John	john@gmail.com	900001234	New York
CUST002	Alice	alice@gmail.com	9123456780	Miami
CUST003	Michael	mike@gmail.com	987651111	Chicago

2. Update - Set-where

This is used to update the content of a record in a relation

Syntax: Update relation name set Field name 1 = data,
field name 2 = data, where field name = data.

Ex:

UPDATE customer set Name='Raha', E-mail = raha@gmail.com
WHERE SSN='CUST001';

After updating

SSN	Name	Email	Phone	Address
CUST001	Raha	raha@gmail.com	98000001234	New York
CUST002	Aliu	aliu@gmail.com	9123456780	Miami
CUST003	Michael	mike@gmail.com	987651111	Chicago

3- Delete from

This is used to delete all records of a relation but it will retain the structure of that relation

a) Delete from: This is used to delete all the records of relation

Syntax: Delete from table-name

Ex: Delete from customer.

After deleting

SSN	Name	Email	Phone	Address

b) Delete from where: This is used to delete a selected record from a relation

Syntax: Delete from relation-name where condition;

Ex: Delete from customer where name='Michael'

After Deleting

SSN	Name	Email	Phone	Address
CUST001	Raha	raha@gmail.com	9800001234	New York
CUST002	Aliu	aliu@gmail.com	9123456780	Miami

5- Truncate

This command will remove the data permanently. But Structure will not be removed

Syntax: Truncate Table (Table name)

Example: Truncate Table customer

After Truncate

SSN	Name	Email	Phone	Address

Queries

1- Retrieve a member name starts with 'R'

Query: Select name from customer where name like 'R%';

Output: Name

Raha

2- List of Accounts where fee between 5000 and 10000

Query: Select * from account where fee between 5000 and 10000;

Output

Name	customer number	Fee	category
Rashmi	29838	5000	Twin room

3- Finding records who has minimum fee

Syntax: Select min(fee) from customer-account;

Output: min fee

Adwaita

4. Finding Records who has fee = 5000

Select * from customer-account where fee = 5000;

Name	customer number	fee	category
Reshmi	29838	5000	Twin room
Ram	29743	9000	Quad room

5. Distinct

Syntax: Select distinct category from customer-account;

Output: category

Twin Room

Quad Room

Triple Room

6. Union

Select name from customer union select name from

Hotel-account

Output:

Name
Reshmi
Advaita
John

VELTECH	
Sl. No.	3-1
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
BY VOICE (5)	5
RECORD (5)	15
TOTAL (20)	1818
DATE	

1818

RESULT: The implementation of DML commands using clauses has been completed successfully

Task 3.2

Aggregate Functions

AIM:

To study and implement aggregate functions (count(), sum(), avg(), min(), max())

PROCEDURE:

Create a table named customer_account

Insert sample records

Write queries using aggregate function

Observe and record the output

commands with explanation

1. count the total number of customer

Select count * AS total_amount from customer_account;

Output: Total amount

4

2. Find the highest fee in the account

Select max(fee) AS highest_amount from customer_account;

Output:

Highest Fee
9700

3. Find the average fee of the customer

Select avg(fee) AS Average_amount from customer_account

Output: 6000

4. Find minimum amount of account

Select min(fee) as min_amount from customer_account

Output: 1800

5. Find the total amount of fee in the customer_account;

Select category, sum(fee) as total_amount from bank_account
group by category;

category	Total amount
Twin Room	5000
Triple Room	4200
Quad Room	7200

6- Find the average fee per category ordered by average fee
 Select category, avg(fee) as Avg-fee from customer-account
 group by category order by avg-fee

category	Avg-amount
Twin Room	3000
Triple Room	5000
Quad Room	6000

VELTECH	
EX No.	3.2
PERFORMANCE (5)	5
RESULT AND ANALYSIS	5
WIVA VOCE (5)	5
RECORD (5)	1
TOTAL (20)	15
AVG MARK	2518

2518

RESULT: The implementation of aggregate functions is executed successfully